

SYSTEM REQUIREMENTS SPECIFICATION (SRS)

MINESHIELD: SMART MINE SAFETY
& MONITORING SYSTEM

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Section 1: Introducing Mineshield, the development team, and the proposed product.

The table below lists all the members who will be working on this product, along with their respective roles. Collaborators will be working on different sections, assisting the executive members. The selected group and product name is Mineshield.

<i>Name</i>	<i>Student Number</i>	<i>Assigned role</i>	<i>Engineering Discipline</i>
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3. CHIRAMBA, AE	225138247	Collaborator	Mechanical
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6. ANDREAS, CV	224007351	Collaborator	Mechanical
7. NEGONGA, T	225059606	Collaborator	Electrical
8. ABRAHAM, PS	225044803	Collaborator	Metallurgical
9. KLIM, GK	225112353	Lead Developer	Mechanical
10. HILOKUAH, HS	222178906	Collaborator	Electronics and Computer
11. HANGHOME, QL	225075040	Collaborator	Electrical
12. ANGULA, FT	224070312	Firestore Lead	Metallurgical
13. SIMASIKU, SAM	225075504	Collaborator	Electrical
14. ESTEVANHU, SI	225077175	Collaborator	Civil
15. GABRIEL, E	224110411	Collaborator	Metallurgical
16. NEHALE, RN	224052993	GitHub Manager	Mining
17. HAIMBONDI, MT	225108860	Collaborator	Agriculture
18. JOSEPHATH, LM	225087383	Collaborator	Mechanical
19. IYAMBO, T	225034328	UI/UX Lead	Electronics and Computer

1.1 Purpose of this document and a brief on the proposed product.

The product in development is called Mine Shield. A mobile application that aims to facilitate the reporting, locating, and detecting of hazards in a mine while simultaneously monitoring the user's well-being. The purpose of this document is to establish standards, provide information for future reference, and set clear objectives, ensuring the efficient progression of the product. All features implemented on the finished product must trace back to the contents of this document, this includes the system design, interaction between react native and Firebase, etc.

1.2 Product Scope

Active features of the proposed product:

- Real-time hazard reporting.
- Monitoring of worker safety with the use of smartphone sensors.
- Provide a live dashboard for supervisors.
- Send emergency alerts.
- Store and analyze safety data.
- Provide the last known location of the user.
- Live map with the ability to illustrate safe and risk zones.

Features that the proposed product will not possess:

- Real-time underground GPS tracking.
- Replace certified mining safety equipment.
- Operate fully offline (limited buffering only).

1.3 Target Engineering Domain:

Mining Engineering and Occupational Safety Engineering.

1.4 Problem Statement:

Mining industries face serious risks such as falls, unsafe zones, and delayed hazard reporting. However, existing technology rely on manual processes and expensive hardware, reducing efficiency and response time when a hazard emerges. Moreover, underground conditions restrict GPS usage and reduce visibility. To address these challenges, Mineshield provides a cost-effective, real-time mobile safety system using smartphone technology and cloud services.

Section 2: System Overview.

2.1 System Description

Mineshield is an app that transforms cellular devices into an intelligent safety tool. It utilizes existing sensors built into mobile devices to detect abnormal movements and sounds, and facilitates the reporting of hazards by creating a platform that enables employees to upload pictures and written descriptions of an observed risk. This information is delivered to the responsible department in real time, preventing injuries and further damage.

2.2 Target Users

<i>User Type</i>	<i>Description</i>	<i>Technical Level</i>	<i>Environment</i>
Mine Workers.	Report hazards, receive alerts.	Low–Medium	Underground / Surface
Supervisors	Monitor safety dashboards	Medium	Control rooms
Safety Officers	Analyze compliance data	Medium–High	Offices
Engineers	Monitor operational risks	High	Technical Environments
Visitors	Receive alerts	Low	Temporary mine access

2.3 Technology Overview

- **Frontend:** Expo + React Native
- **Backend:** Firebase
 - Firebase Authentication
 - Cloud Firestore
 - Firebase Storage
 - Firebase Cloud Messaging (FCM)

2.4 Firebase Data Model

Collections and Fields

Users	Hazards	Alerts	Zones	Analytics	Sensor Logs
user ID	hazard ID	alert ID	zone ID	Recorded ID	Log ID
name	user ID	user ID	zone name	date	User ID
email	description	type	risk level	Total hazards	Noise level
role	image URL	location		Hazard type	Motion Data
assigned Zone	location	timestamp			timestamp
	timestamp				
	status				

2.5 Assumptions and Constraints

- Android 10+ required
- Internet required for real-time sync
- GPS unavailable underground
- Smartphone sensors must function
- Firebase services must be used within its limitations
- Worker should be in possession of mobile device at all times for safety monitoring
- Users are aware of the limitations, working conditions and how to navigate the application.
- Supervisors have stable internet connection and are logged to the app.
- Mobile device has sufficient battery capacity to be operational for the intended work period.

Section 3: Functional Requirements.

ID	Requirement Statement	Priority	Firebase Service
FR-001	The system shall allow User registration using email and password.	High	Firebase Authentication
FR-002	The system shall allow secure login and logout	High	Firebase Authentication
FR-003	The system shall allow users to report hazards with description and image	High	Firestore + Storage
FR-004	The system shall store hazard reports in real time	High	Firestore
FR-005	The system shall display hazards on a live map	High	Firestore
FR-006	The system shall Classify zones by risk level.	High	Firestore
FR-007	The system shall detect Falls using accelerometer data.	High	Firestore
FR-008	The system shall Monitor noise levels.	Medium	Firestore
FR-009	The system shall send alerts when thresholds are exceeded.	High	Firebase Cloud Messaging
FR-010	The system shall send SOS alerts with last known location.	High	Firestore
FR-011	The system shall display a supervisor dashboard.	High	Firestore
FR-012	The system shall allow viewing of past hazards.	Medium	Firestore
FR-013	The system shall generate analytics reports.	Medium	Firestore
FR-014	The system shall allow visitors to receive alerts.	High	Firestore
FR-015	The system shall support image uploads.	Medium	Storage

Section 4: Non-functional Requirements.

Performance

- Dashboard loads within 3 seconds.
- Real-time updates within 2 seconds.
- Map will update 5 seconds after report has been made.

Security

- Authentication required for all access.
- Role-based access control.
- Notifications that originate from personal sources shall be blocked.
- Visitors can only use the app under the supervision of an employee after admin approval.

Usability

- Simple and intuitive interface.
- Easy registration process.
- Video tutorials to guide visitors and staff.

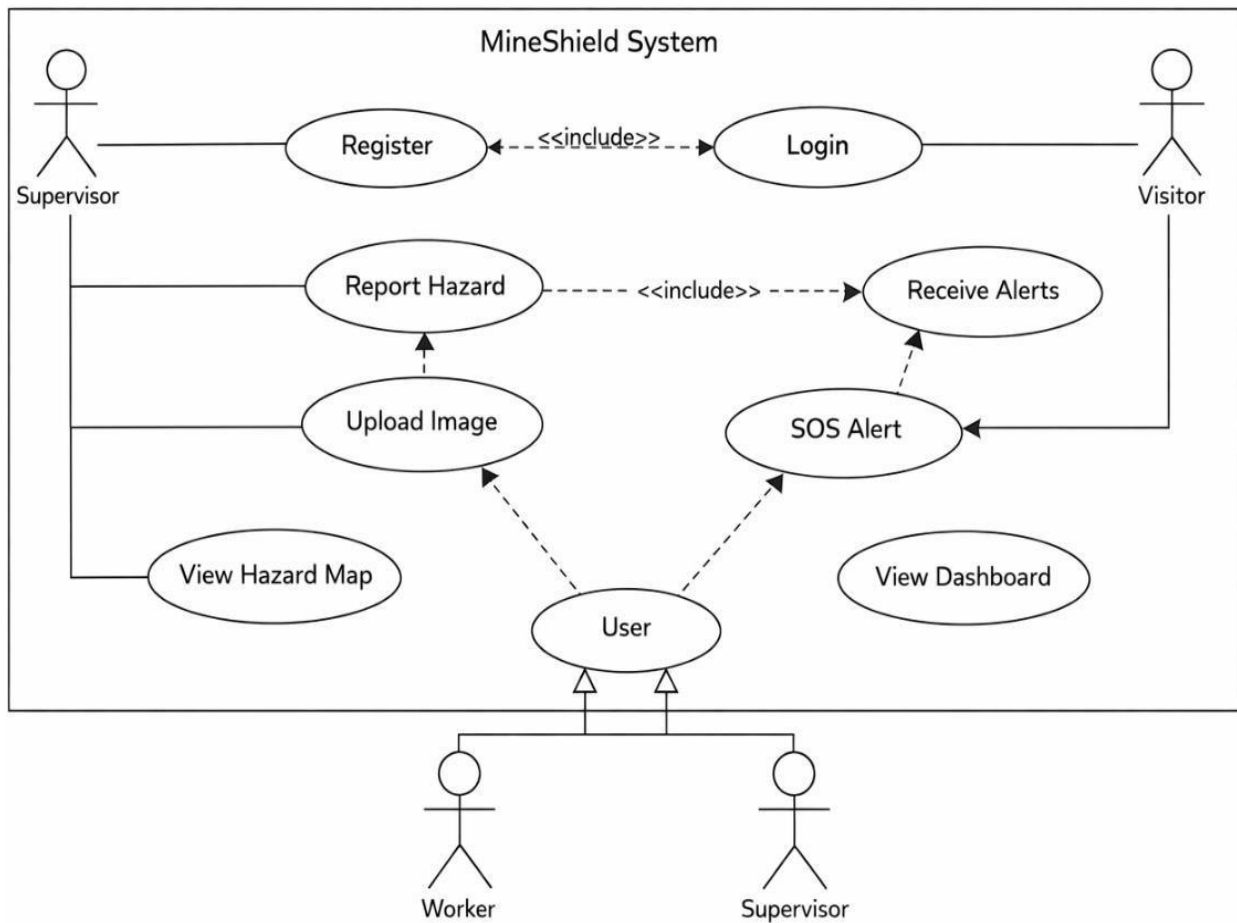
Compatibility

- Android 10+ devices.
- Expo Go support.

Reliability

- Data not lost during network failure.
- Offline buffering supported.
- Abnormal movements result in the display of a modal window, confirming user wellbeing.

Section 5: Case Diagram illustrating product interaction with users.



Case 1: Report Hazard

- **Actor:** Worker / Visitor
- **Description:**
User logs in, reports hazard with description and image. System stores data and updates dashboard.
- **Outcome:** Hazard is visible to all users.

Case 2: Fall Detection Alert

- **Actor:** System
- **Description:**
System detects abnormal motion, triggers alert, and sends notification with last known location.
- **Outcome:** Supervisor is notified immediately.

Use Case 3: Supervisor Monitoring

- **Actor:** Supervisor
- **Description:** Supervisor logs in and monitors real-time hazards and alerts via dashboard.
- **Outcome:** Quick decision-making, reducing the response time.

Section 6: Conclusion.

Mineshield provides a cost-effective, real-time safety solution for mining environments. By integrating hazard reporting, sensor monitoring, and cloud-based analytics, the system improves hazard awareness, emergency response, and overall safety. But just like any other security system, it only makes mining operation safer and should not be expected to be a main security feature. Supervisors should ensure they have a stable internet connection and that there is always personal monitoring the dashboards to ensure that reports and alerts are responded quickly. During visits, registered staff will take responsibility for a certain number of visitors and write their details in their accounts.